

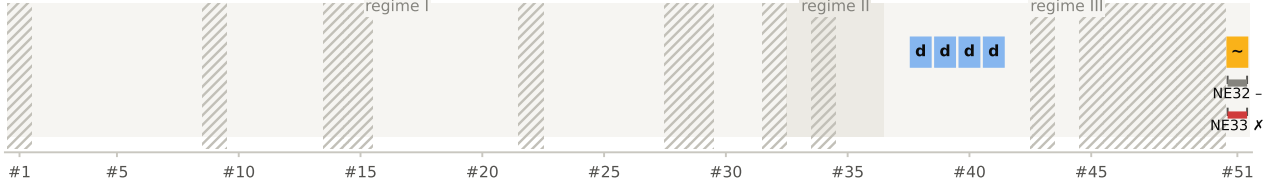
H98: #51 is a random-field system

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1.5/5 Usefulness 1/5

Bottom line. #51's static structure is a random field with a weak pull, but its dynamics are not independent agents in a field: rivals co-move mostly through conversation chosen by niche, and some days move together.

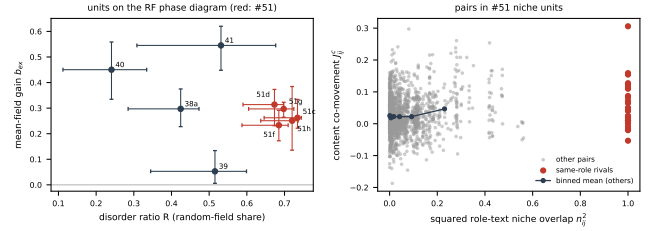
level: hypothesis · pipeline checks 86% · updated 2026-10-04



Goal periods tested (1 failed, 1 mixed, 4 descriptive, 1 n/a). ✓ supported ✗ failed mixed d descriptive ... pending. Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

H22 refuted a spin glass in #51 and described a random-field system instead. H98 fits that picture as a mean-field random-field $O(n)$ model on 9 non-holdout regime-III units: 5 private-role #51 units and 4 shared-goal weeks (#38–#41). After style removal, 67–73% of the static content configuration in #51 is agent-specific (disorder ratio R), against 24–53% in shared-goal weeks. A weak positive pull acts on top ($b_{ex} = 0.23\text{--}0.31$). Same-role rivals co-move because their role texts overlap: the niche slope fitted on non-rivals predicts the rival excess (bge). But reads and replies absorb 70% of that slope, and day-to-day overlaps are synchronized in 2 of 5 units. **The claim fails by its pre-registered rule;** the static random-field part stands. The holdout is not run.



Observables. Left: each unit on the random-field phase diagram (disorder ratio R , agent bootstrap CI; mean-field gain b_{ex} , day-bootstrap CI). #51 units (red) sit at higher R than every shared-goal week. Right: content co-movement against squared role-text overlap for all pairs in the 7 #51 niche units. Rivals (red) have identical role texts ($n^2 = 1$). The binned mean of non-rivals rises with n^2 .

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator who assigns private roles, the role text sets only 5–14% of where an agent's content sits. Most of the field is the agent's own projects, and a newcomer takes days to settle into its role (day-1 own-role percentile 0.57 vs 0.83 for incumbents). Agents with overlapping roles read and answer each other, and that conversation, not a shared field, carries most of their co-movement. To keep two roles from converging, give them non-overlapping role texts and separate rooms. In the room-split native, one moved agent lost all pull toward #general ($0.42 \rightarrow 0.01$). As a monitor, R measured after style removal tells a private-goal swarm ($R \approx 0.7$) from a shared-goal one ($R \leq 0.53$).

PHYSICAL MODEL

Mean-field random-field $O(n)$ model (models 01, 11). The spin s_{id} is agent i 's whitened, style-residualized content on day d ($n = 32$). The quenched field h_i holds role, own projects and priors; $\hat{r}_i$ is the whitened role text. J pulls toward the room mean; g_d is the day field. Within a day, window fluctuations follow a shared drive η_w filtered by each agent's niche projector P_i , so co-movement grows with the squared niche overlap n_{ij}^2 .

$$s_{id} \propto h_i + J m_{-i,d} + g_d + z_{id} + \epsilon_{id}, \quad R = \frac{\Delta^2}{\Delta^2 + M^2}$$

$$x_{iw} = (\alpha P_i + \beta) \eta_w + \xi_{iw} \Rightarrow \mathbb{E} J_{ij}^c \propto \alpha^2 n_{ij}^2 + 2\alpha\beta + n\beta^2$$

$$\hat{T}_{SR} = \beta_n (\overline{n^2}_{SR} - \overline{n^2}_U), \quad G = T_{SR} - \hat{T}_{SR}$$

H98-random-field-51 · generated 2026-10-07 17:50 by in-fra/summaries/build_summaries.py · card hypotheses/H98-random-field-51/README.md · ratings: H98 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
P1 $R(\#51) > \text{every contrast unit}$	0.67–0.73 vs 0.24–0.53 (gte same; supported not without style removal)	supported
P2 $b_{ex} > 0$, all < 0.5	5/5 #51 CI > 0 (0.23–0.31); #41 0.55	mixed
P3 no collective switching (W)	BC < 0.555 in 5/5; W 3.05 (p failed (2/5) 0.007), 1.61 (p 0.048) in 51d, 51g	failed
P4 niche slope predicts rival excess	β_n 0.064 [0.009, 0.118]; G -0.034 bge only [−0.095, 0.028]; gte β_n 0.010 (n.s.)	bge only
P5 reads+replies keep $\geq 70\%$ of β_n	0.084 $\rightarrow$ 0.025 (30%)	failed
P6 stance ferromagnetic	mean 0.51–0.64; β_s 0.019 [−0.066, 0.103]	supported
P7 role text in the field	R_{role}^2 0.05–0.14, $p \leq 0.001$ (7/7)	supported
NE32 isolated newcomers	0–1 statements while isolated	untestable
#focus split	pull 0.42 $\rightarrow$ 0.01 (one mover), unchanged (other)	mixed
NE33 joiners on day 1	own-role percentile 0.57 vs 0.83	failed

GOAL PERIODS IN DETAIL

G38 d descriptive
G39 d descriptive
G40 d descriptive
G41 d descriptive
G51 $\sim$ mixed
NE33 $\times$ failed
NE32 – n/a

FAITHFULNESS SCORECARD

A mapping 1 F identifiability 1
B assumptions 1 G ground truth 1
C adequacy 1 H comparative 1
D unfitted predictions 1 I transfer 0
E interventional 1

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The niche result holds in bge only; #51 is the least model-robust period (DQ5). $\hat{T}_{SR}$ extrapolates from non-rival overlaps $n^2 \leq 0.56$ to rivals' $n^2 = 1$. DQ2 reply parents are partly content-selected, so the reply control may over-adjust; reads alone remove about 30%. R depends on the reference centre (Amendment A0) and on style residualization. b_{ex} includes any common window drive. The sources of the W excess (a join day, a slow drift) are post hoc readings. Two of three natives failed or were untestable; the room split has two agents.

ROUND 2

Model reading as niche-gated coupling $J_{ij} = J_0 f(n_{ij}^2)$ and test co-movement given reads. Add a day-level amplitude field to the overlap model. Fit newcomers' settling time into their role field.

- **Essence.** how much of a mixed-motive swarm's structure is set by assigned roles (field), and how much by who reads whom (coupling).
- **H98-R1** Niche-selected coupling.
- **H98-R2** Day-level amplitude field.
- **H98-R3** Settling time.

Second figure. Niche slope β_n per #51 unit and random-effects pool (diamonds), with same-lab control only (circles) and with ledger reads and DQ2 replies added (squares). In bge the pooled slope falls from 0.084 to 0.025. In gte (dark) it is null from the start.

